NIMCET PREP PRO — ENTERPRISE GUIDE

# Complete Architecture, API Setup, Permissions & .env Configuration Guide

Tailored for AIR 1 Preparation | 10-Yr NIMCET + 10-Yr CAT Reasoning | Full MERN Stack

| | | |
|---|---|---|
| **Target Exam:** NIMCET 2025/2026 | **Target NIT:** NIT Trichy (#1 Ranked) | **Stack:** React 18, Vite, Node, Express, Mongo |
| **AI Model:** Google Gemini Flash | **Auth:** JWT + Google OAuth 2.0 | **Status:** Fully Standalone (In-Memory + DB) |

## 1. Executive Overview & Core Features

NIMCET Prep Pro is an advanced, high-performance web platform designed specifically for students aiming for **All India Rank 1 (AIR 1)** in the NIT MCA Common Entrance Test (NIMCET). The application bridges the gap between weak mathematical foundations and top-tier competitive mastery through line-by-line step explanations, prerequisite breakdowns, and high-difficulty CAT reasoning sets.

- **Comprehensive 10-Year Question Bank:** Complete real papers from NIMCET 2015 through 2024 with categorized difficulty (Easy, Moderate, Hard).
- **10-Year CAT DILR & Reasoning Sets:** Advanced seating arrangement, logical deductions, syllogisms, and coding problems to build unmatched reasoning speed.
- **Interactive Chapter & Topic Navigation:** Collapsible sidebar organizing Coordinate Geometry, Calculus, Algebra, Probability, Vectors, and Trigonometry.
- **Gemini AI Step-by-Step Problem Solver:** When a student fails or requests assistance, Gemini provides calculus breakdown, exact formulas used, and foundational concepts to learn first.
- **Authenticated Student Profile & Real-Time Persistence:** Dynamic dashboard tracking student accuracy, score benchmarks, goal NIT, and personalized study notes.

## 2. Project Architecture & File Structure

The project follows clean MVC architecture on the backend and modern React modular components on the frontend:

## 3. Required External APIs & Step-by-Step Setup

To run the application with all advanced capabilities (AI step tutor and Google Sign-In), you will need two standard free API keys. The system includes built-in fallback modes if keys are not provided immediately.

### A. Google Gemini AI API (Required for AI Tutor)

1. **Visit Google AI Studio:** Go to https://aistudio.google.com/app/apikey.
2. **Generate API Key:** Sign in with your Google account and click *Create API Key*.
3. **Free Tier Limits:** Google provides generous free tier credits suitable for thousands of student prompt queries daily.
4. **Add to .env:** Paste your key in server/.env under GEMINI_API_KEY=AIzaSy....

### B. Google OAuth 2.0 Client ID (Required for Google Sign-In)

1. **Open Google Cloud Console:** Visit https://console.cloud.google.com/apis/credentials.
2. **Configure OAuth Consent Screen:** Navigate to *APIs & Services* → *OAuth consent screen*. Select **External**, name your application *NIMCET Prep Pro*, and provide your email address.
3. **Create Credentials:** Click + *Create Credentials* → *OAuth client ID*.
4. **Set Application Type:** Choose **Web application**.
5. **Authorized JavaScript Origins:** Add http://localhost:3000 (and your deployment domain).
6. **Authorized Redirect URIs:** Add http://localhost:3000.
7. **Copy Client ID:** Paste in both client/.env (as VITE_GOOGLE_CLIENT_ID) and server/.env (as GOOGLE_CLIENT_ID).

## 4. System Permissions & Port Requirements

| Resource / Port | Permission Required | Reason / Purpose |
|---|---|---|
| `Port 5000` | Local Inbound Binding | Express REST API server listener |
| `Port 3000` | Local Inbound Binding | Vite development server for React UI |
| `Outbound HTTPS` | Internet Access (Port 443) | Calling Google Gemini API and Google OAuth token verification |
| `Port 27017 (Optional)` | Local / Remote TCP | MongoDB connection (Memory store activates automatically if offline) |
| `Local Disk / File` | Read / Write User Directory | Storing cached questions and local study assets |

## 5. Complete .env File Reference (Ready to Copy)

Below are the complete, production-ready environment files configured for local and network operation:

### Backend Configuration: server/.env

```
# Server Network Port PORT=5000 NODE_ENV=development # Database Connection (MongoDB Atlas Cloud Cluster or Local) MONGO_URI=mongodb+srv://:@cluster0.6qcjvxi.mongodb.net/nimcet_prep?retryWrites=true&w;=majority&appName;=NIMCET # Authentication Security Key (Replace with any random 32+ char secret string) JWT_SECRET=nimcet_air_1_super_secret_jwt_key_2025_elite JWT_EXPIRE=30d # Google Gemini AI API Key (Obtain from https://aistudio.google.com/app/apikey) GEMINI_API_KEY=AIzaSy...your_gemini_api_key_here # Google OAuth 2.0 Client ID (From https://console.cloud.google.com) GOOGLE_CLIENT_ID=your_client_id.apps.googleusercontent.com
```

**Resolved: Windows DNS SRV Error (querySrv ECONNREFUSED):**
On Windows, local router/ISP DNS servers often block DNS SRV lookups (_mongodb._tcp...). The backend has been engineered with an automated Google Public DNS resolver (dns.setServers(['8.8.8.8', '8.8.4.4', '1.1.1.1'])) directly inside server/config/db.js, automatically solving this connection error and enabling instant Atlas synchronization.

### Frontend Configuration: client/.env

```
# Backend API Target Base URL VITE_API_URL=http://localhost:5000/api # Google OAuth 2.0 Web Client ID (Must match the one authorized for http://localhost:3000) VITE_GOOGLE_CLIENT_ID=your_client_id.apps.googleusercontent.com
```

## 6. User Authentication & Profile Customization Flow

As specifically required, the navbar dynamically updates based on session state. Here is how user information and authentication are managed:

- **Navbar Dynamic Replacement:** Before login, *Log In* and *Sign Up* buttons are visible. Once authenticated (via Email/Password or Google), these buttons are instantly removed and replaced with the user's interactive **Profile**

**Avatar & Rank Badge**.

- **Direct Redirection:** Clicking the avatar redirects directly to /profile, where all student academic records are displayed.
- **Academic & Goal Customization:** Students can view and edit their *Target NIMCET Year* (2025, 2026, 2027), *Dream NIT Destination* (NIT Trichy, Surathkal, Warangal, etc.), *Math Foundation Level* (Weak to AIR 1 Elite), *Phone Number*, *Current College*, and *Personal Study Strategy*.
- **Dynamic In-Memory & MongoDB Persistence:** Edits are immediately saved via PUT /api/auth/profile, ensuring data updates are retained dynamically throughout the study session.

## 7. Quick-Start Execution Commands

```
# 1. Open Terminal in the project root: cd "C:\Users\Vinayak Gupta\Desktop\my code\nimcet-prep-pro" # 2. Start the Backend API Server: cd server npm start # Runs on http://localhost:5000 # 3. Open a second Terminal for the Frontend Client: cd ..\client npm run dev # Runs on http://localhost:3000
```

**AIR 1 Preparation Tip:** For maximum performance in NIMCET, start each session by solving 10 Calculus questions on Easy/Moderate difficulty, then switch to CAT Reasoning sets to train rapid pattern recognition under timed mock conditions.